

EXECUTIVE PRESENTATION

WALMART DEMAND FORECASTING & INVENTORY OPTIMIZATION

Walmart Store Sales | 45 Stores | 81 Departments | 2010–2012

420K

Records

45

Stores

14.30%

MAPE

THE OPERATIONAL PROBLEM

OVERSTOCK

Excess inventory accumulates, storage costs rise, markdowns used to clear stock that should never have been ordered.

 Deadstock ·  Margin erosion ·  Storage cost

UNDERSTOCK

Shelves run empty, sales lost permanently. A customer who can't find what they need does not wait.

 Lost sales ·  Churn ·  Zero revenue recovery

"The data exists. What doesn't exist is a consistent way to translate it into decisions managers can actually plan around."

THE DATASET AT A GLANCE

420,285

Records

45

Stores

81

Departments

2.5 Yrs

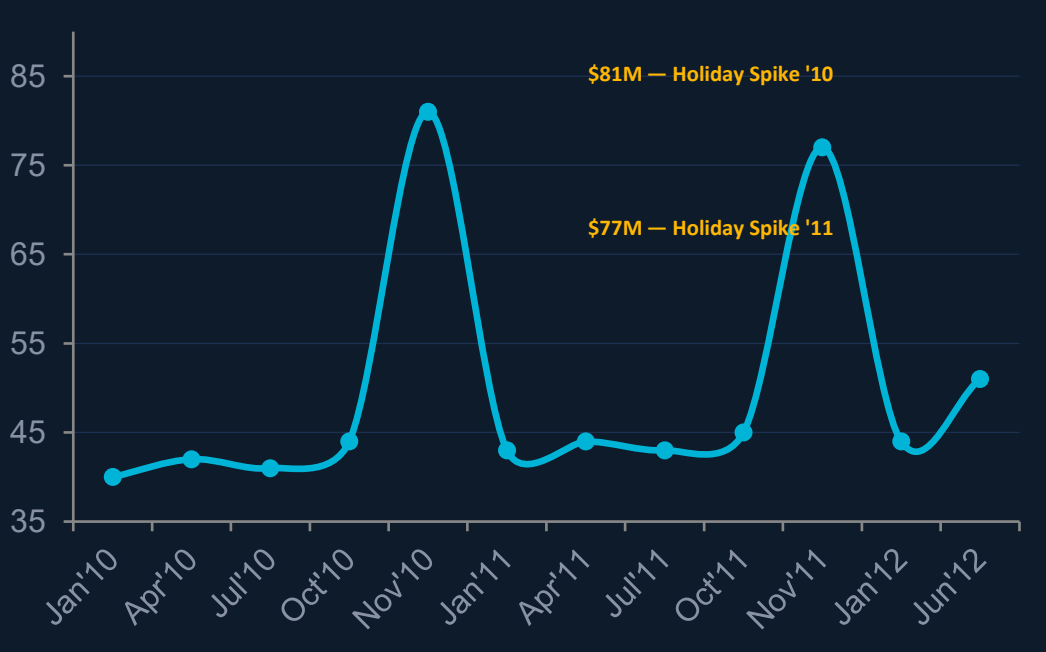
Time Span

DATA MERGE

train.csv + features.csv — weekly sales records merged with external conditions: temperature, CPI, unemployment, fuel price, and promotional markdown data.

"Holiday weeks behave so differently from the rest of the year they are effectively a separate business problem."

WHAT THE DATA REVEALS: SALES STRUCTURE



Top Store **Store 20**

\$301M Total Sales

Top Dept **Dept 92**

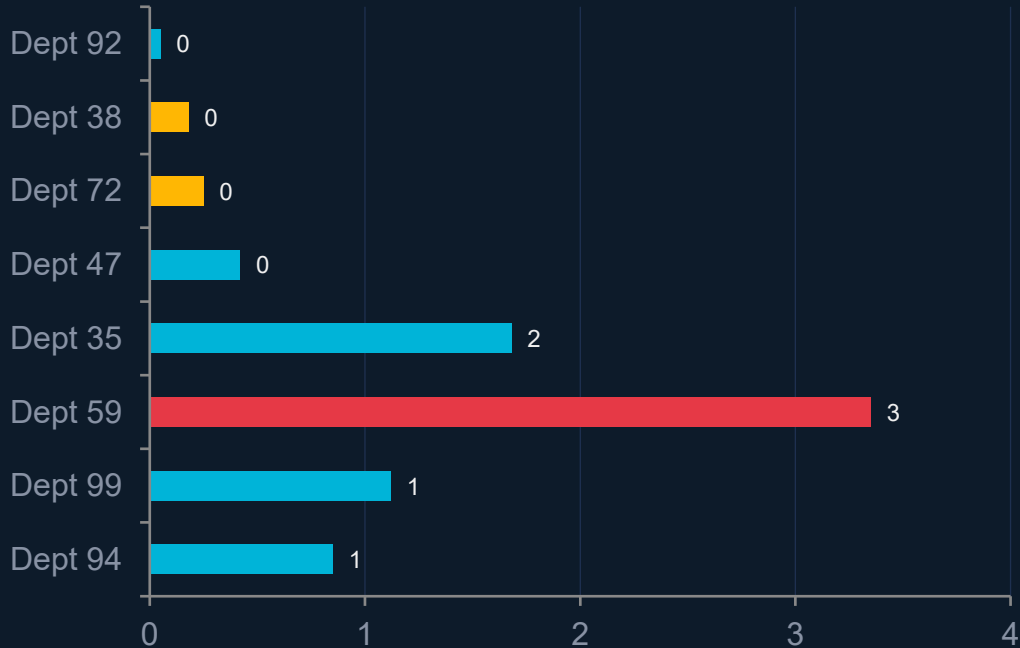
\$484M Total Sales

Holiday Lift **+7.5%**

vs. regular weeks

"Revenue is highly concentrated — top 3 stores and top 3 departments carry disproportionate financial risk."

VOLATILITY: NOT ALL DEMAND IS EQUAL



Most Volatile

CV 3.35

Store 35, Dept 59

Stable Floor

CV ≤ 0.05

Core grocery depts

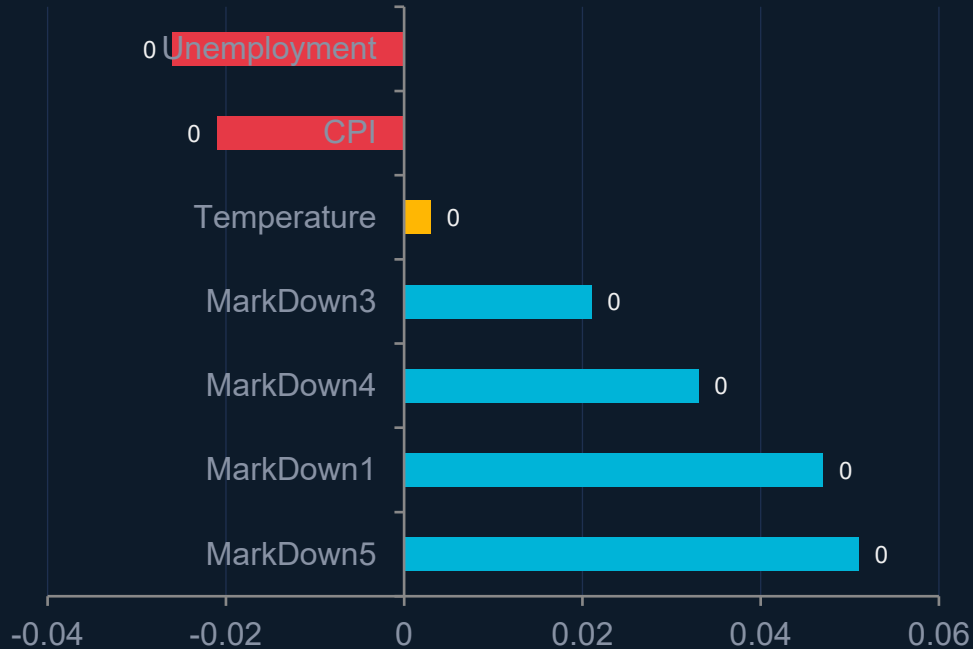
Volatile Cluster

Depts 59, 99, 94

Category-level risk

"Demand volatility is not random — it clusters at the category level, and safety stock decisions must reflect that."

DO EXTERNAL FACTORS DRIVE SALES? (THE ANSWER SURPRISES)



THE KEY FINDING

Promotional markdowns move sales.
Macroeconomic conditions and weather do not.
This simplifies the forecasting problem significantly.

- ✓ Markdowns → Signal
- ✗ Macro / Weather → Noise

"External signal is weak — the demand story is internal, not macro."

THE FIRST MODEL FAILED. THAT WAS THE FINDING.

SARIMA

BASELINE MODEL

MAE

\$34,263

MAPE

190.04%

"Flat forecast where actuals surge. No mechanism for holiday behaviour."

PROPHET

CHAMPION MODEL

MAE

\$2,558

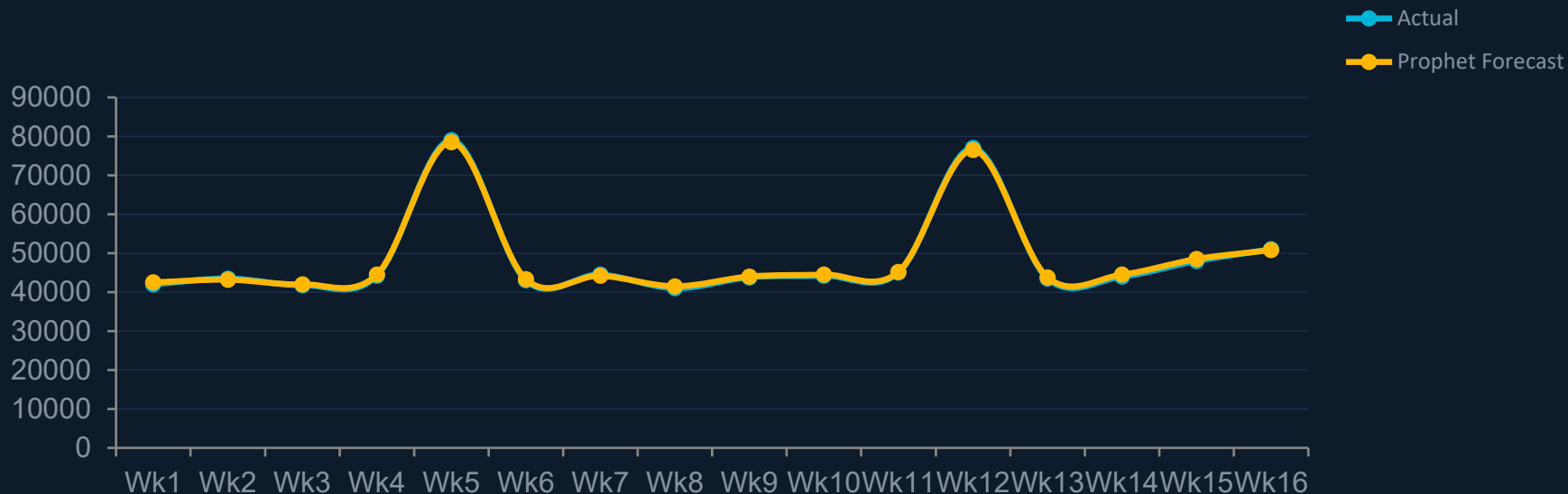
MAPE

14.30%

"Holiday regressor absorbs the spikes SARIMA treats as noise."

"A 190% MAPE is not a modelling failure — it is proof that holiday weeks are a structurally different business problem."

THE FORECAST: WHAT DEMAND LOOKS LIKE NEXT QUARTER



MAE \$2,558

RMSE \$4,871

MAPE 14.30%

"Prophet captures both baseline demand and holiday spikes with meaningful accuracy — ready for operational use."

FROM FORECAST TO INVENTORY DECISION



"Every number is auditable. Change the assumptions, the model recalculates — no black box."

ASSUMPTIONS

$S = \$50 \cdot h = 25\% \cdot C = \$10 \cdot \text{Lead Time} = 1 \text{ wk} \cdot \text{Service Level} = 95\%$

EOQ → Safety Stock → Reorder Point — From forecast to decision in a single auditable chain.

WHAT OPERATIONS SHOULD DO NEXT

01 Holiday Weeks as Separate Planning Regime

Treat Nov–Dec as a distinct operational mode. The 7.5% average lift and \$81M spike cannot be handled by standard forecasting parameters.

02 Tier Inventory by Store Revenue

A 6× revenue gap exists between top and bottom stores. Safety stock allocation should reflect that asymmetry — not be applied uniformly.

03 Safety Stock by Volatility Cluster

CV ranges from 3.35 (extreme) to ≤ 0.05 (stable). Blanket safety stock wastes capital at the stable floor and creates risk at the volatile extreme.

04 Markdown Planning Anchored to Top Depts

Depts 92, 95, and 38 account for \$1.33B combined. Promotional calendar and markdown decisions should start here — not be applied chain-wide.

"The data says where the risk is. These four decisions are how operations responds to that signal."

"The model doesn't replace the operations manager. It tells them where to look, when to act, and how much to order."



Demand Has Structure

The seasonal pattern repeats predictably.
It can be planned around.



Volatility Is Identifiable

High-risk combinations are known. Safety
stock can be targeted.



The Decisions Are Quantifiable

EOQ, safety stock, reorder point — from
data, not intuition.